Using these back in Eq. (27.4.1), the Lagrangian density becomes

$$\begin{aligned}
\mathscr{L} = & -\sum_n (D_\mu\phi)^*_n (D^\mu\phi)_n \\
& -\frac{1}{2}\sum_n \left(\overline{\psi_{nL}}\,\gamma^\mu (D_\mu\psi_L)_n\right) + \frac{1}{2}\sum_n \left(\overline{(D_\mu\psi_L)_n}\,\gamma^\mu\,\psi_{nL}\right) \\
& -\frac{1}{2}\sum_{nm} \frac{\partial^2 f(\phi)}{\partial\phi_n\,\partial\phi_m}\left(\psi^{\rm T}_{nL}\epsilon\,\psi_{mL}\right) - \frac{1}{2}\sum_{nm}\left(\frac{\partial^2 f(\phi)}{\partial\phi_n\,\partial\phi_m}\right)^* \left(\psi^{\rm T}_{nL}\epsilon\,\psi_{mL}\right)^* \\
& -\sum_n \left|\frac{\partial f(\phi)}{\partial\phi_n}\right|^2 \\
& +i\sqrt{2}\sum_{Anm}\left(\overline{\psi_{nL}}\,(t_A)_{nm}\,\lambda_A\right)\phi_m - i\sqrt{2}\sum_{Anm}\phi^*_n\left(\overline{\lambda_A}\,(t_A)_{nm}\,\psi_{mL}\right) \\
& -\frac{1}{2}\sum_A\left(\xi_A + \sum_{nm}\phi^*_n\,(t_A)_{nm}\,\phi_m\right)^2 - \frac{1}{4}\sum_A f_{A\mu\nu}f_A^{\mu\nu} \\
& -\frac{1}{2}\sum_A\left(\overline{\lambda_A}\,(\not{D}\lambda)_A\right) + \frac{g^2\theta}{64\pi^2}\epsilon_{\mu\nu\rho\sigma}\sum_A f_A^{\mu\nu}f_A^{\rho\sigma}\,.
\end{aligned} \tag{27.4.8}$$

Lorentz invariance requires the fields ψ_{nL}, λ_A, and $f_{A\mu\nu}$ to have vanishing vacuum expectation values, while the tree-value vacuum expectation values of the ϕ_n are at the minimum of the potential

$$V(\phi) = \sum_n\left|\frac{\partial f(\phi)}{\partial\phi_n}\right|^2 + \frac{1}{2}\sum_A\left(\xi_A + \sum_{nm}\phi^*_n\,(t_A)_{nm}\,\phi_m\right)^2 . \tag{27.4.9}$$

This potential is positive, so *if* there is a set of field values at which $V(\phi)$ vanishes, then this is automatically also a minimum of the potential. For $V(\phi)$ to vanish at some field value $\phi_n = \phi_{n0}$, it is necessary and sufficient that

$$\mathscr{F}_{n0} = -\left[\frac{\partial f(\phi)}{\partial\phi_n}\right]^*_{\phi=\phi_0} = 0 \tag{27.4.10}$$

and

$$D_{A0} = \xi_A + \sum_{nm}\phi^*_{n0}\,(t_A)_{nm}\,\phi_{m0} = 0\,. \tag{27.4.11}$$

This in turn is the necessary and sufficient condition for supersymmetry not to be spontaneously broken, since Eq. (26.3.15) gives $\langle\delta\psi_{nL}\rangle_{\rm VAC} = \sqrt{2}\langle\mathscr{F}_n\rangle_{\rm VAC}\,\alpha_L$, and Eq. (26.2.16) gives $\langle\delta\lambda_A\rangle_{\rm VAC} = i\langle D_A\rangle_{\rm VAC}\,\gamma_5\alpha$.

It is worth stressing here that spontaneous symmetry breaking is more difficult for supersymmetry than for other symmetries. For most symmetries of the action there will be field configurations at which the symmetry is unbroken and the potential is stationary, but the symmetry will